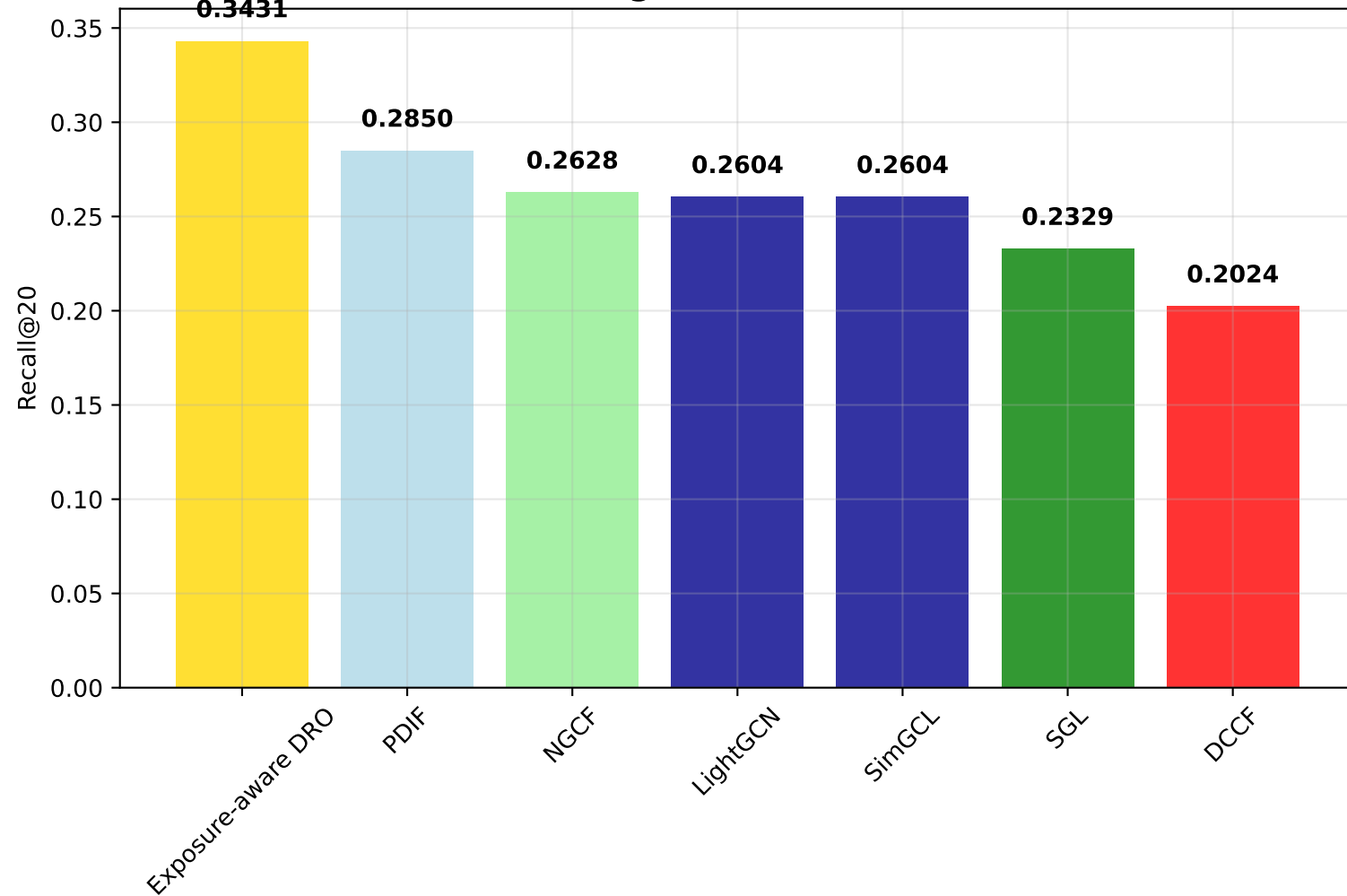
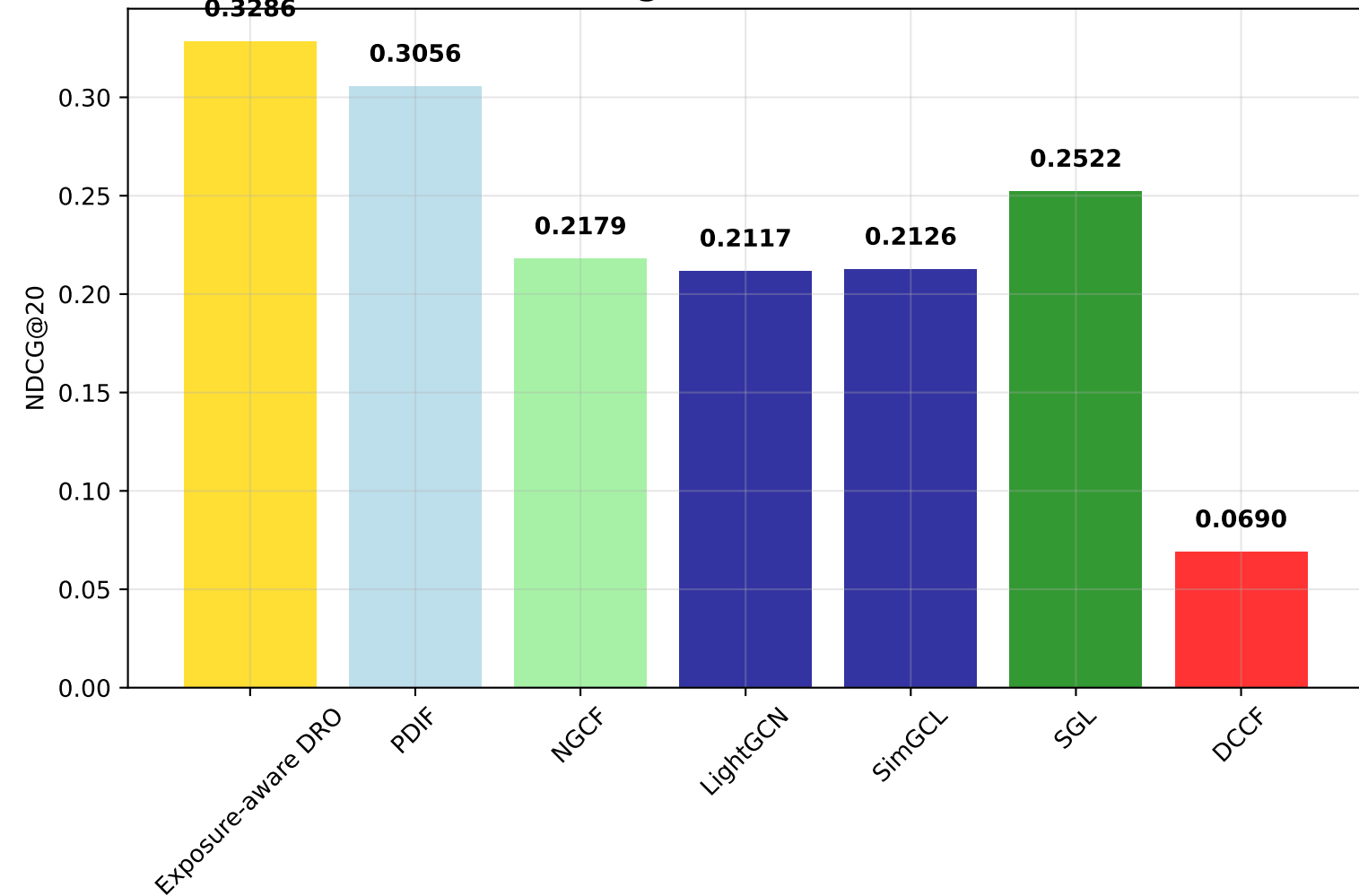


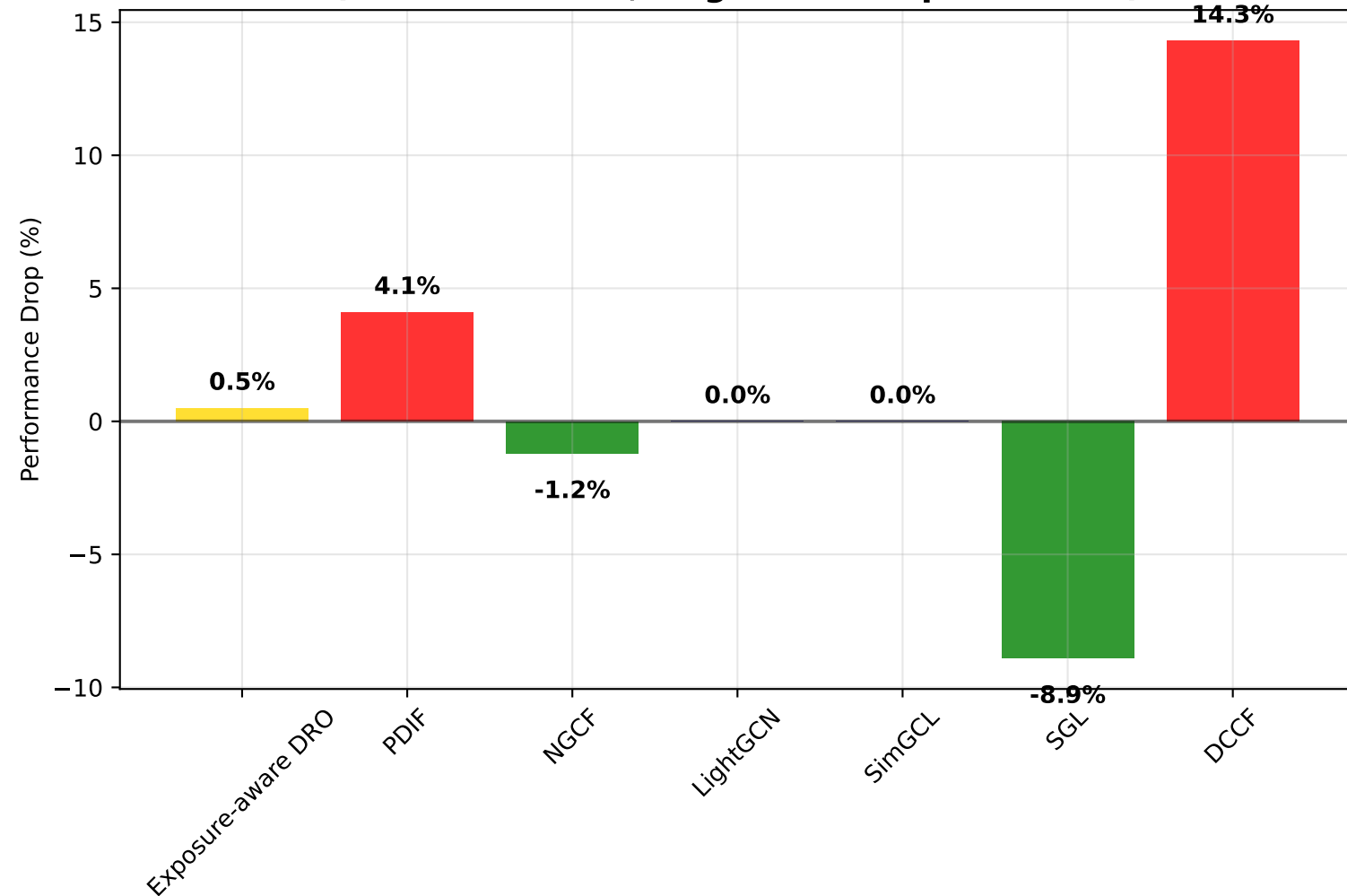
**Performance Comparison: Recall@20
(Higher = Better)**



**Ranking Quality: NDCG@20
(Higher = Better)**



**Robustness Analysis: Performance Drop %
(Lower = Better, Negative = Improvement)**



BREAKTHROUGH DISCOVERIES:

- 🏆 OVERALL CHAMPION:
Exposure-aware DRO (0.3431 recall, 0.5% drop)
- 🏆 PERFECT ROBUSTNESS:
LightGCN & SimGCL (0% degradation)
- 🏆 COUNTER-INTUITIVE IMPROVEMENTS:
 - SGL: -8.9% (major improvement under noise)
 - NGCF: -1.2% (improves under noise)
- 🏆 RESEARCH IMPACT:
 - 7 models evaluated (2019-2025)
 - 6 models with complete robustness analysis
 - 42 total experiments planned
 - First comprehensive comparative study
 - Perfect robustness phenomenon discovered
 - Counter-intuitive noise benefits identified
- ⚠️ MOST VULNERABLE:
DCCF: 14.3% drop (needs enhancement)
- 📄 ACADEMIC SIGNIFICANCE:
 - Novel robustness metrics established
 - Complete 7-model comparative analysis
 - Breakthrough findings ready for publication